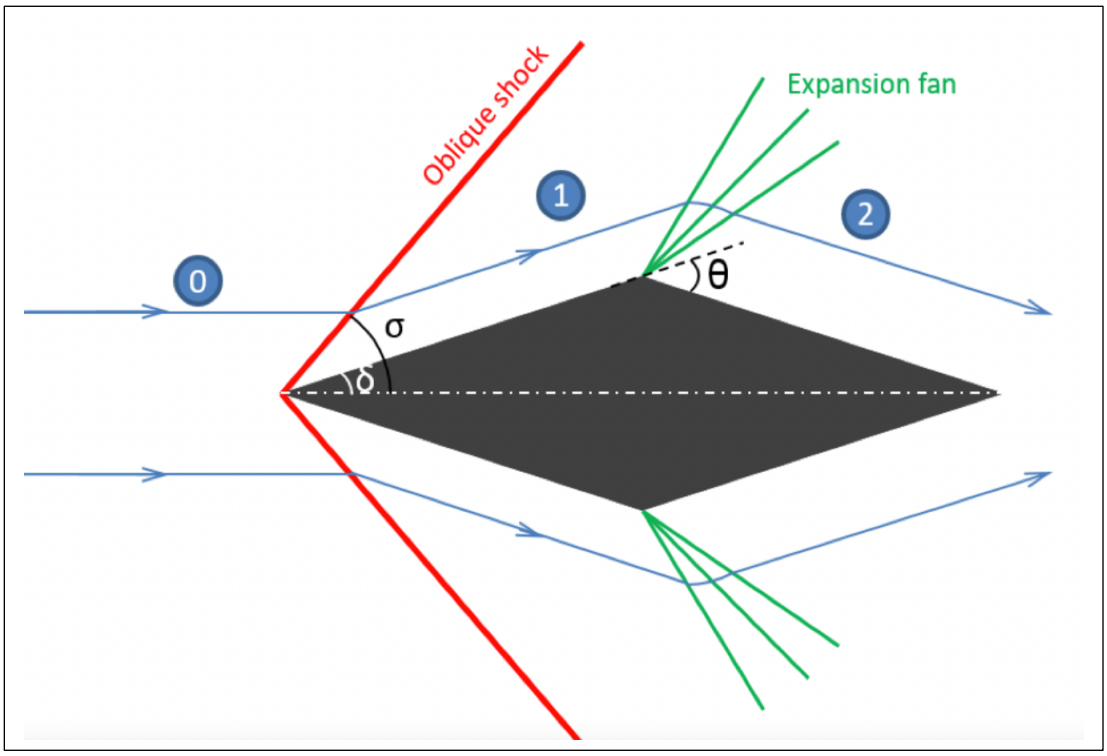


SUPERSONIC AIRFOIL DESIGN

Submitted By

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1. Problem Statement

The objective of this project is to design a supersonic airfoil with a maximum lift to drag ratio under given geometric parameters and given design aerodynamic constraints. Also, the oblique shock and Prandtl-Meyer relations are solved in MATLAB to compute the lift coefficient (cl), wave drag coefficient (cd), and pitching moment coefficient (Cm_LE) around the leading edge of a supersonic airfoil using the two-dimensional shock-expansion wave method. The purpose of the project is to solve a design optimization challenge in which specific parameters (design variables) must be determined to get the highest demonstrable performance under given constraints.

2. Airfoil Geometric Model

A supersonic airfoil is required for a new supersonic aircraft design, as shown in Fig. 1. The leading and trailing edges of the airfoil must be sharp. As shown in Fig. 2, the upper and lower surfaces of the airfoil are approximated by two straight panels in this project.

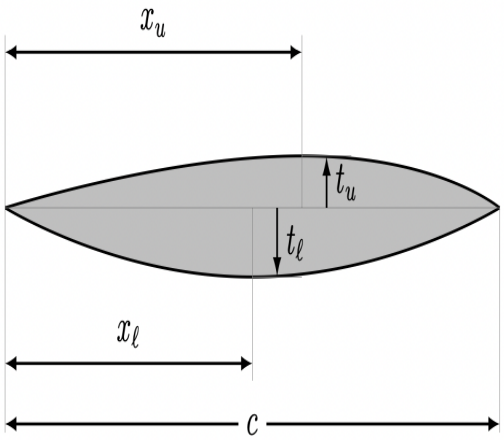


Figure 1: Supersonic Airfoil

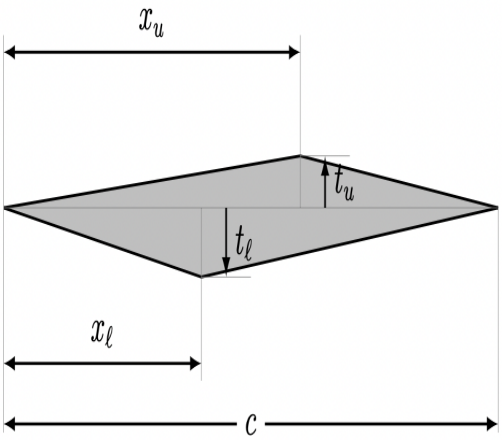


Figure 2: Double Wedge Airfoil

The design Mach number is 2.5. And the control variables for the design are angle of attack, xu, xl, tu, and tl. Here xu, xl, tu, and tl are the geometric control parameters.

xu = Position for the maximum thickness of top panel
xl. = Position for the maximum thickness of bottom panel
tu = Maximum thickness of top panels
tl = Maximum thickness of bottom panels

The design constraints in this project are:

- Max. thickness

$$\frac{t_u}{c} + \frac{t_\ell}{c} \geq 0.05$$

- Pitching moment about the leading edge

$$|c_{m,LE}| \leq 0.08$$

- Lift coefficient

$$c_\ell \geq 0.2$$

Because of the design constraints for maximum thickness, we can eliminate one of the control variables i.e., tl.

3. Aerodynamic Performance Analysis

Theory about formation of shock wave/ expansion fan:

For a given angle of attack, the diamond shape airfoil produces both expansion and an oblique shockwave. When the wave is turned away from itself, the upper surface of the leading edge produces an expansion wave i.e., when the angle of attack exceeds the angle of inclination of the upper panel with the chord. When the wave is turned into itself, the upper surface of the leading edge produces an oblique shock wave i.e., when the angle of attack is less the angle of inclination of the upper panel with the chord. However, when the angle of attack is equal to the angle of inclination of the upper panel, no shock or expansion wave is seen at the leading edge of upper panel. The lower surface of the leading edge, on the other hand, has an oblique wave as the wave compresses. In diamond shape airfoil, there is always an expansion fan at the maximum thickness edges. If the angle of attack has negative value, the upper and lower surface behaves in a opposite way.

Analysis of pressure difference:

As we know that the pressure increases across the oblique shock wave and decreases across the expansion fan. The variation of pressure difference between the upper and lower surface is responsible for lift force. Hence, we prefer expansion fan on the leading edge of upper panel (decrease in pressure) and an oblique shock wave on the leading edge of lower panel (increase in pressure). The drag against the flow is provided by the thickness of the airfoil. The drag force is the outcome of this. The

pitching moment is caused by variations in the resultant force at the front and back surfaces. The two-dimensional chock-expansion wave theory is used to calculate the lift, drag, and pitching moment coefficients. MATLAB code has been written to solve it.

Effects of angle of attack and airfoil geometric parameters on c_l, c_d, and Cm_LE.

a.) Angle of attack: For a specific airfoil if the angle of attack is greater, both lift and induced drag coefficients are greater. However, if the critical angle of attack is exceeded then the lift coefficient decrease and the drag coefficient increase which results in stalling the aircraft.

b.) xu and xl: Keeping xu towards the trailing edge gives less inclination of upper panel and for small angle of attack we can have expansion fan in the leading edge of the upper panel which reduces the pressure. However, if you keep xu towards the leading edge the angle of attack must be increased to have an expansion fan. Similarly, keeping xl towards the leading edge provide strong oblique shock which further increases the pressure in lower surface and provide a higher lift. However, keeping xl too close to leading edge results in greater pressure difference between the front panel and results in stalling.

c.) tu and tl: The thickness of the airfoil is responsible for the increment in drag coefficient. So, for positive angle of attack tu is kept thicker than tu and for negative angle of attack tl is thicker than tu. The lower panel thickness is responsible for higher pressure in lower section. If the difference between pressure is not as desired drag increases and the aircraft results into stalling.

4. Design Optimization

The MATLAB code was written to solve the two-dimensional shock wave expansion theory. For different angle of attack, xu, xl, tu, and tl the results (cl, cd, cm_LE) was produced and copied to the spreadsheet. The code not only solved for the diamond shaped airfoil but also for the triangular shaped airfoil and produced results without the design constraints. So, from the spreadsheet only the data that met the design constraints were taken. The results of MATLAB that met the design constraints is tabulated in the results section below.

5. Results and Discussion

All the data shown below are calculated for the value of chord length =1.

AOA	xu	xl	tu	tl	Max Thickness	cm_LE	cd	cl	cl/cd ratio
5	0.7	0.5	0.06	0.12	0.18	-0.0577	0.0948	0.2185	2.3056
5	0.7	0.5	0.06	0.11	0.17	-0.0575	0.0829	0.2055	2.4784
5	0.7	0.5	0.05	0.12	0.17	-0.0578	0.0925	0.2238	2.4206
5	0.8	0.5	0.05	0.12	0.17	-0.0543	0.0931	0.221	2.373
5	0.8	0.4	0.05	0.12	0.17	-0.0497	0.1053	0.2296	2.1807
5	0.9	0.3	0.05	0.13	0.18	-0.0431	0.1593	0.276	1.7326
5	0.8	0.3	0.04	0.12	0.16	-0.0508	0.1311	0.2618	1.997
5	0.8	0.3	0.04	0.1	0.14	-0.0511	0.0925	0.2225	2.405
5	0.85	0.7	0.05	0.11	0.155	-0.0672	0.0726	0.2038	2.8056
5	0.6	0.48	0.05	0.1	0.15	-0.0587	0.0707	0.2006	2.8382
6	0.6	0.48	0.05	0.1	0.15	-0.0722	0.0795	0.2332	2.9332
6	0.85	0.65	0.05	0.1	0.15	-0.0751	0.0763	0.2277	2.9836
6	0.8	0.3	0.04	0.11	0.15	-0.0652	0.1221	0.2761	2.2614
6	0.7	0.6	0.05	0.1	0.15	-0.0772	0.0756	0.2312	3.0597
6	0.6	0.45	0.04	0.11	0.15	-0.0696	0.0922	0.2514	2.7272
6	0.5	0.5	0.04	0.11	0.15	-0.0727	0.0882	0.2489	2.8229
6	0.4	0.6	0.04	0.08	0.12	-0.0789	0.0562	0.2151	3.8292
6	0.4	0.6	0.03	0.09	0.12	-0.0769	0.063	0.2283	3.6233
6	0.4	0.6	0.04	0.1	0.14	-0.0793	0.0741	0.236	3.1863
6	0.9	0.6	0.03	0.07	0.1	-0.0713	0.0481	0.2048	4.2604
7	0.97	0.6	0.03	0.07	0.1	-0.0788	0.0567	0.2305	4.0659
7	0.8	0.6	0.092	0.09	0.18	-0.08	0.0829	0.2198	2.6519
7	0.9	0.6	0.092	0.09	0.18	-0.0676	0.082	0.2101	2.5621
7	0.85	0.35	0.1	0.08	0.18	-0.0662	0.0874	0.2082	2.382
7	0.8	0.35	0.035	0.13	0.16	-0.0784	0.1492	0.3274	2.195
7	0.9	0.35	0.035	0.13	0.16	-0.0742	0.1499	0.3236	2.1588
7	0.9	0.4	0.04	0.12	0.16	-0.0731	0.1274	0.2983	2.3415
7	0.8	0.4	0.04	0.1	0.14	-0.0792	0.0957	0.2727	2.8509
7	0.1	0.75	0.03	0.12	0.15	-0.0733	0.1031	0.2056	1.9947
7	0.6	0.4	0.025	0.12	0.14	-0.0795	0.1159	0.3015	2.601
8	0.9	0.9	0.09	0.09	0.18	-0.1336	0.1045	0.2862	2.7377
8	0.9	0.5	0.09	0.09	0.18	-0.0789	0.095	0.2486	2.6169
8	0.85	0.4	0.1	0.1	0.2	-0.0786	0.118	0.2641	2.2382
8	0.85	0.45	0.1	0.1	0.2	-0.0796	0.1127	0.2597	2.3036

8	0.83	0.39	0.11	0.14	0.25	-0.0779	0.1999	0.334	1.6711
8	0.8	0.4	0.111	0.12	0.235	-0.08	0.1627	0.2993	1.8398
8	0.6	0.4	0.09	0.14	0.23	-0.0548	0.1869	0.2443	1.3067
8	0.62	0.4	0.09	0.14	0.23	-0.0538	0.1867	0.2439	1.3062
8	0.7	0.4	0.1	0.13	0.23	-0.0455	0.1681	0.2087	1.2412
8	0.9	0.4	0.09	0.15	0.24	-0.0763	0.2175	0.3663	1.3067
9	0.9	0.4	0.14	0.08	0.22	-0.0784	0.1088	0.2359	2.1675
9	0.95	0.4	0.14	0.08	0.22	-0.0721	0.1072	0.2313	2.1588
9	0.95	0.45	0.14	0.08	0.22	-0.072	0.104	0.2286	2.1972
9	0.95	0.4	0.15	0.07	0.22	-0.0713	0.0981	0.2118	2.1593
9	0.97	0.35	0.1	0.12	0.22	-0.0797	0.1788	0.3382	1.8911
9	0.6	0.4	0.12	0.15	0.27	-0.0641	0.2541	0.2787	1.097
9	0.65	0.4	0.12	0.13	0.25	-0.0557	0.1921	0.215	1.1194
9	0.7	0.4	0.15	0.15	0.3	-0.0429	0.268	0.2337	0.872
9	0.8	0.4	0.15	0.15	0.3	-0.0248	0.2633	0.2236	0.8492
9	0.85	0.45	0.14	0.16	0.3	-0.02	0.2543	0.2253	0.8862
10	0.95	0.4	0.16	0.12	0.28	-0.078	0.1895	0.3149	1.6618
10	0.95	0.45	0.15	0.13	0.28	-0.08	0.1967	0.3318	1.6872
10	0.95	0.45	0.16	0.1	0.26	-0.0792	0.1451	0.2737	1.8863
10	0.75	0.45	0.16	0.16	0.32	-0.0449	0.2964	0.2537	0.8559
10	0.8	0.45	0.16	0.16	0.32	-0.0344	0.2933	0.2479	0.8451
10	0.85	0.45	0.16	0.16	0.32	-0.0221	0.2901	0.2407	0.8296
10	0.8	0.4	0.15	0.14	0.29	-0.0351	0.2488	0.2235	0.8985
10	0.9	0.5	0.16	0.17	0.33	-0.0106	0.2872	0.2302	0.8016
10	0.95	0.35	0.165	0.13	0.29	-0.0794	0.2312	0.3541	1.5314
10	0.6	0.23	0.12	0.08	0.2	-0.08	0.156	0.2053	1.3164
10	0.65	0.23	0.12	0.08	0.2	-0.0769	0.1545	0.2041	1.3212

In the above tabulated results, the feasible design not only had expansion fan on the leading edge of the upper section but also, had the oblique shock wave. The ratio of lift coefficient and drag coefficient is less than 1 for the oblique shock wave in the leading edge. The small change in the geometric parameter has a significant impact on the results. For higher xu the design constraints are met.

Plot of the best designed airfoil

